

Name: _____

Collaborators: _____

THAYER SCHOOL OF ENGINEERING • DARTMOUTH

ENGS 28, W26

LAB 1 — ALL KINDS OF VARIATIONS ON BLINKY

Due: Tuesday January 13, 2026 2:15 PM

Contents

1 Overview	2
1.1 Place in course	2
1.2 Learning Objectives	2
1.3 Equipment	2
2 Prelab	2
3 Technical Study	2
3.1 LED Properties	2
3.2 LED Measurements	3
4 Design Challenges	7
5 Postlab	9

Deliverables

1 Prelab Quiz (5 points - counted separately)	2
2 LED Properties (3 points)	3
3 LED Measurements (3 points)	5
4 Current Limiting Resistor (2 points)	5
5 Light Intensity (2 points)	6
6 BlinkSIM (2 points)	8
7 BlinkSEQ (3 points)	8
8 BlinkCNT (5 points)	8

1 Overview

1.1 Place in course

In class we wrote our first bit of code to blink an LED. This lab will solidify that learning by writing a number of variations on the blinky theme. This lab will also give you additional practice using Ohm's Law to calculate voltages and currents. You must read the supplemental handout available on Canvas (see Deliverable 1).

1.2 Learning Objectives

- Gain experience reading data sheets.
- Become proficient with multi-meter measurements of voltage, current, and resistance.
- Design and prototype an embedded system with LED outputs.
- Practice writing clear, correct, and well-documented C code.

1.3 Equipment

- Nucleo + Breadboard
- 3 LEDs (Mix of Red and Green)
- 3 220Ω resistors¹
- 1 $1k\Omega$ resistor
- Digital Multimeter (DMM)

2 Prelab

Review the readings for this week (see the daily pages). Note the checkpoint questions in the readings from the ENGS 31 eBook. Read through the supplementary material.

Deliverable 1: Prelab Quiz (*5 points - counted separately*)

The problems in the supplementary material must be turned in on Canvas with the Prelab 1 Quiz.

3 Technical Study

3.1 LED Properties

On Canvas, go to the assignment page for this lab, and find the datasheets for the Kingbright LEDs you will be using this term. There are separate datasheets for the red LED and the green LED. Pick one and answer the following questions:

¹Your kit may have 200Ω resistors, these are fine too

Deliverable 2: LED Properties (3 points)

Using values from the datasheet, answer the following questions.

- What is the typical forward voltage V_f for your diode?

Answer: _____

- What is the maximum forward current I_f for your diode?

Answer: _____

- What material is your LED composed of?

Answer: _____

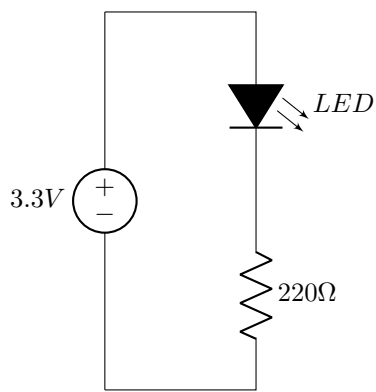
- Is the color of your LED light emitted determined solely by the color of the plastic housing? Explain your reasoning in the space below.

- Assume V_{OH} (the output high voltage) for the Nucleo is V_{CC} (3.3V). Using I_F and V_F from above, what is the smallest resistor that will keep the current safe for your LED? Show your work. (Note: refer to Figure 1a if you are unsure what the circuit should look like).

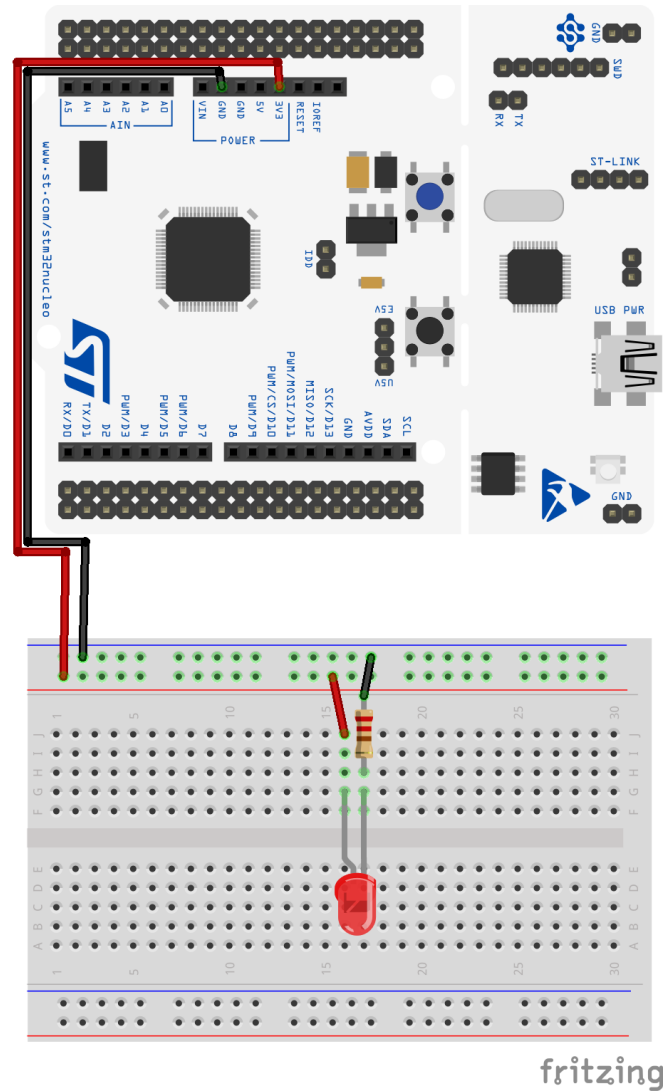
3.2 LED Measurements

With your Nucleo unplugged, select an LED and a 220Ω resistor from your kit. Construct the circuit shown in Figure 1a.

- The anode of the LED will be connected to the red 3.3V power rail on the breadboard, also called V_{CC} . Always use **RED** for wiring connections to this power rail.
- Connect the cathode of the LED to one leg of the 220Ω resistor.
- Connect the other leg of the 220Ω resistor to the circuit ground reference (0V), also called GND. Always use **BLACK** for wiring connections to ground.
- Connect the **RED** (V_{CC}) rail to the 3.3V pin in the *Power* section of the Nucleo board.
- Connect the **Black** (*Gnd*) rail to the *GND* pin in the *Power* section of the Nucleo board.



(a) Schematic View



(b) Sample Breadboard Layout

Figure 1: Initial LED Circuit

Deliverable 3: LED Measurements (3 points)

Using your voltmeter, make the measurements described below. Your measurements should agree quite well with Kirchoff's voltage law ($V_{CC} = V_F + V_R$). If they don't, think about what you might have done wrong (reversing the positive and negative probes is a common error - in this circuit, all your measured voltages should be positive). Keep at it until you have a good result, then write down your three voltages.

- V_{CC} - the voltage across the power rails.

Answer: _____

- V_F - the voltage across the diode.

Answer: _____

- V_R - the voltage across the resistor.

Answer: _____

Ohm's law states that the voltage across a resistor is proportional to the current flowing through the resistor multiplied by its resistance, or $V = IR$.

Deliverable 4: Current Limiting Resistor (2 points)

Using the voltage across the resistor you measured in Deliverable 3 answer the following questions

- Calculate the current flowing through the resistor (I_R).

Answer: _____

- How much current flows through the LED?

Answer: _____

- In the space below, compare this current to the maximum indicated by your LED's datasheet. Is the current limiting resistor doing its job? Why or why not?

Replace the 220Ω resistor with a $1k\Omega$ resistor and repeat the measurements from before.

- V_{CC} - the voltage across the power rails.

Answer: _____

- V_F - the voltage across the diode.

Answer: _____

- V_R - the voltage across the resistor.

Answer: _____

Deliverable 5: Light Intensity (*2 points*)

In the space below, answer these questions.

1. Did the forward voltage drop across your LED change?
2. Did the voltage drop across the resistor change? If so, what does this tell us about the circuit?
3. How did the intensity of the light created by the LED change? Why?

4 Design Challenges

There are three closely related design challenges in this week's lab. Begin by wiring up three LEDs (any color), with three 220Ω resistors, to Port A, bits 5, 6 and 7.

Figure 2b shows a photo of the layout on the breadboard, alongside a schematic diagram of the wiring for the LEDs.

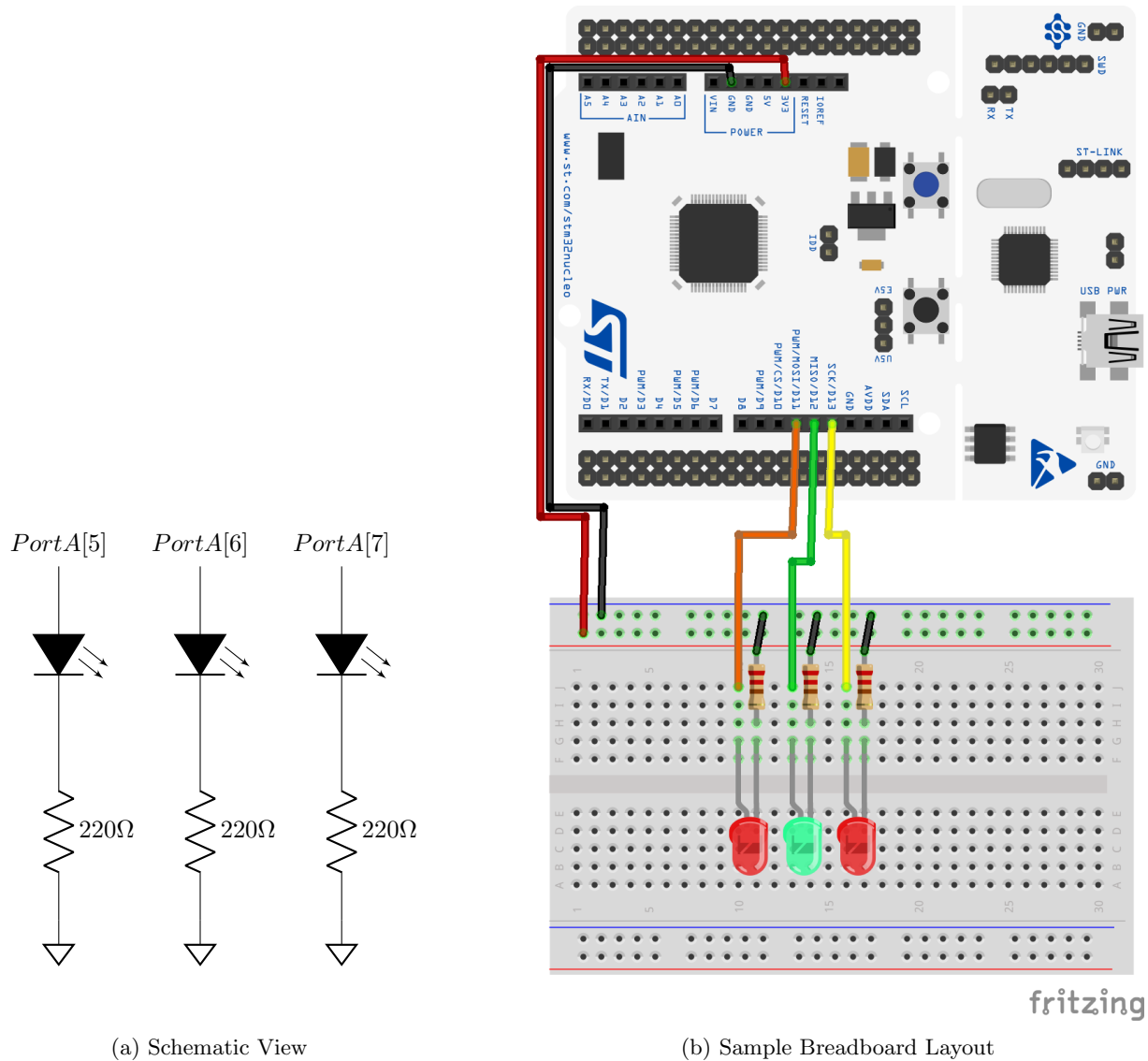


Figure 2: Initial LED Circuit

After wiring it up, design, implement, and test each of these modified blinky designs. The first two were introduced in Wednesday's class.

1. blinkySIM: Flash all three LEDs simultaneously.
2. blinkySEQ: Flash the three LEDs in sequence, one at a time: 0 (PA5), 1 (PA6), 2 (PA7), 0, etc

3. blinkyCNT: Design and implement a three-bit counter (0 to 7), displaying the three bits with the three LEDs. This is harder. You can use a `uint8_t` variable to represent the count, but you have to figure out how to isolate bits 0, 1, and 2 so they can be displayed on the LEDs (consider a bit mask). Make sure the counter counts slowly enough that you can easily see the lights flash in binary sequence: 000, 001, 010, ..., 111, 000, *etc*

Use your best C programming style. Be sure to clearly document your code with inline comments and a descriptive header. Make a copy of the template project from class to get started.

As you work through these problems, think about the build and test philosophy described in class and how it applies to this system. Also think about how you are organizing your file system so you can refer to these three programs at a later date with ease.

Demonstrate each of your designs to a TA or instructor. If you are unable to demonstrate the design to a TA or instructor, upload a video of each challenge working to Canvas.

Deliverable 6: BlinkSIM (2 points)

Obtain a signature or submit a video of blinkySim. Upload your code to Canvas.

TA or Instructor Signature _____

Deliverable 7: BlinkSEQ (3 points)

Obtain a signature or submit a video of blinkySeq. Upload your code to Canvas.

TA or Instructor Signature _____

Deliverable 8: BlinkCNT (5 points)

Obtain a signature or submit a video of blinkyCnt. Upload your code to Canvas.

TA or Instructor Signature _____

5 Postlab

If you wish, you may choose to “permanently” attach a couple of LEDs with their current limiting resistors to your breadboard. These are very useful and will be used in later labs. Choose an out of the way spot near the narrow edge of your breadboard to place the LEDs. Trim the legs of the LED and the resistor so they fit tightly to the board. Be sure to leave room to connect a jumper so you can control the LED later! Figure 3 shows one possible layout. Feel free to take some of the hookup wire from the cabinet to form jumper wires like those seen in the figure. (Ask for help if you aren’t sure which tools to use!)

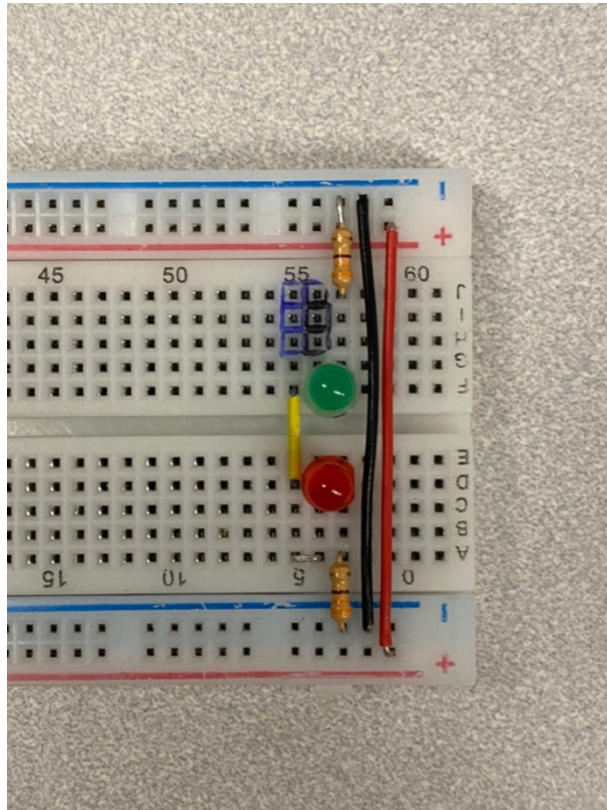


Figure 3: One layout of some "permanent" LED